MILITARY SPECIFICATION

COMPUTER, STALL WARNING CPU-81/A

1. SCOPE

1.1 This specification covers certain design and all performance requirements of a stall warning computer, designated CPU-81/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5583	Transmitter, Position, 28-V DC
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-T-25627	Transmitter, Synchro, Angle of Attack or Side Slip
MIL-C-26500	Connectors, General Purpose, Electrical Miniature, Circular Environment Resisting, 200°C Ambient Temperature

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Expotential Distribution

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MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MS17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS24264	Connectors, Receptacle, Electrical Flange Mount, Miniature
MS33540	Safety Wiring, General Practices for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33638	Cases, Instrument, Flange-Mounted, Aircraft

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Toxic and corrosive fumes. The materials, as installed in the computer and under the service conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.3.4 Nonferrous materials. Nonferrous materials shall be used for all parts except where ferrous materials are essential.

3.3.4.1 Nonferrous materials contained within hermetically sealed inclosures shall be considered suitably protected against corrosion. Requirements specified for fungicidal and corrosion protective treatment and anodizing of aluminum alloy parts shall not be applicable for parts within hermetically sealed inclosures. Steel parts within such inclosures shall be cadmium plated in accordance with QQ-P-416 type I, class C.

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3.3.5 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion from fuel, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.6 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.7 Protective treatment. When materials are used in the construction of the computer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The computer shall be designed to provide computed stall margin signals to the indicator (or indicators) and to provide an activation signal for the stick shaker.

3.4.1 The computer shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.2 Reliability. The computer shall have a mean-time-between-failures (MTBF) of 1,000 hours at a discrimination ratio of 1.5.

3.4.3 Maintainability. The computer shall be so designed that overhaul, including replacement of worn parts, and failed electrical components, adjustments, and calibration may be accomplished by instrument and electronic repairmen trained in the repair of electronic and similar equipment.

3.4.4 Fire hazard. The unit shall be so designed as to safeguard against hazards to the aircraft in the event of malfunction or failure, and the maximum operating temperature of surfaces of any instrument component contacted by combustible fuel or vapor shall not exceed 180°C due to self-heating.

3.4.5 Special tools. The design of the computer assembly shall, to the greatest possible extent, be such as to accommodate disassembly, re-assembly, and service maintenance by those tools and items of maintenance equipment which are normally available to Military maintenance personnel. The requirements of a line analyzer are excluded from this restriction.

3.4.6 Service life. A design objective shall be 3,000 hours of continuous operation under normal service conditions without overhaul.

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3.4.7 Elapsed time meter. A 10,000-hour elapsed time indicator conforming to MS17322 shall be provided to record the electrical activation time of the computer.

3.4.8 Ruggedness. The equipment shall be sufficiently rugged to withstand all conditions incident to handling, shipping, storage, installation, service, and maintenance.

3.4.9 Chassis rigidity. The chassis shall be sufficiently rigid to support the internal components against vibration, microphonics or other environmental conditions.

3.4.10 Compatibility. The computer shall be compatible with the C/KC-135 aircraft, its environment, and shall operate with the following, as specified. Connection of the equipment shall be as shown on figure 1.

a. Angle of attack transmitter conforming to MIL-T-25627

b. Stich shaker

c. Flap position transmitter conforming to MIL-T-5583

d. Airbrake position transmitter

e. Mach transmitter

f. Stall margin indicator.

3.4.10.1 Component tolerances. Component tolerances on gain, phase, friction, backlash, et cetera, shall be established in designing the computer so that they will be compatible with tolerances of the equipment specified in 3.4.10 and so that component selection will not be required in field maintenance during equipment service life.

3.4.11 Self-test capability. The computer shall be of such design that it can be cycled through a self-test function on a preflight basis.

3.5 Performance. The computer shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperatures — operating temperatures ranging from -54° to +71°C and storage temperatures ranging from -65° to +98°C

b. Humidity — relative humidity up to 100 percent

c. Altitude — pressures ranging from 31 inches Hg down to 0.80 inch Hg (approximately 80,000 feet)

d. Salt fog — exposure to simulated salt sea atmosphere for 48 hours

e. Vibration — vibration per MIL-STD-810, curve C

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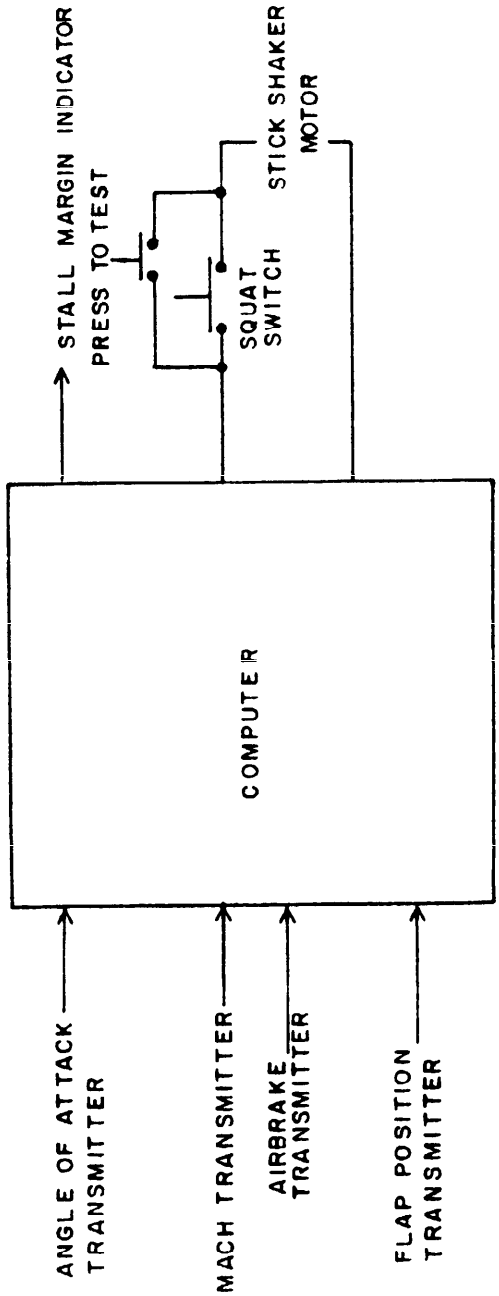


FIGURE 1. System Diagram

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f. Fungus — fungus growth as encountered in tropical climates

g. Sand and dust — sand and dust as encountered in desert areas

h. Thermal shock — temperature changes from +85° to -40°C within 5 minutes

i. Shock — shock force of 15g's for 11 milliseconds

j. Rain — rainfall as encountered in any locale.

3.5.1 Radio noise. Radio noise measurements shall be in accordance with MIL-I-6181 over the frequency range of 150 kc to 150 mc.

3.5.2 Dielectric strength. There shall be no insulation breakdown when 500V ac is applied between isolated pins and between pins and the case for a period of 10 seconds.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists shall be used unless approved by the procuring activity. Electronic tubes shall not be used. The computation network of the computer shall contain no electro-mechanical devices.

3.8 Adjustment devices (potentiometers, linkage adjustment, et cetera). The unit may contain internal adjustments to be used by the vendor for calibration. It shall not be necessary to make adjustments at any time during the life of the unit. All adjustments shall be provided with a positive locking means to prevent any change in adjustments due to the environmental conditions specified herein. Wherever possible, adjustments necessary only for initial flight testing shall be replaced by fixed parameters. A means shall be provided to prevent use of oversize adjusting tools. Positive mechanical stops shall be provided and materials and construction shall be such as to minimize mechanical bind. Adjustments and circuits shall be sufficiently isolated from one another so that the setting of one adjustment will not affect the setting of any other adjustment.

3.9 Electrical connectors. All electrical connectors mating with interconnection harnesses shall conform to the applicable requirements of MIL-C-26500, except where space and weight consideration dictate otherwise. The use of connectors that do not meet the requirements of MIL-C-26500 shall be as approved by the procuring activity. Functionally noninterchangeable connectors shall not be physically interchangeable. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400. Careful attention shall be given to the selection of subunit connectors, if used, to avoid contact trouble.

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3.10 Safetying. All threaded parts shall be positively locked or safetied by safety-wiring, self-locking nuts, or other acceptable methods. Provisions shall be made for safetying or adequately fastening by other means all electric connector mating plugs and pneumatic connections. The use of star washers will not be acceptable. Screws or other components shall be safety wired in accordance with MS33540.

3.11 Transistors. Transistors shall be in accordance with MIL-S-19500.

3.12 Functions. The computer shall be capable of accepting signals from the angle of attack transmitter, flap position transmitter, mach transmitter, and airbrake transmitter and shall provide the following outputs: One output proportional to V_S and capable of driving the stall margin indicator and the other output to activate the stick shaker when the conditions shown on figures 2 and 3 are met.

3.12.1 Computer functions shall be defined in terms of flap position, angle-of-attack synchro angles, and mach. Operational functions are shown on figure 2 and are further defined as follows. The computer shall incorporate provisions for adjusting stick shaker signal and the limiting functions and shall be capable of adding signal bias corresponding to figure 3 when airbrakes are activated.

a. The 0° flap curve shown on figure 2 consists of a straight line from 12.3° α wing to mach 0.35; a straight line connecting 12.3 α and 0.35 mach to 10.6° α and 0.6 mach; a straight line from 10.6° α to 0.7 mach

b. The 20° flap curve shown on figure 2 consists of a straight line from 11.4° α wing to mach 0.35; a straight line connecting 11.4° α and 0.35 mach to 9.3° α and 0.6 mach; a straight line from 9.3° α to 0.7 mach

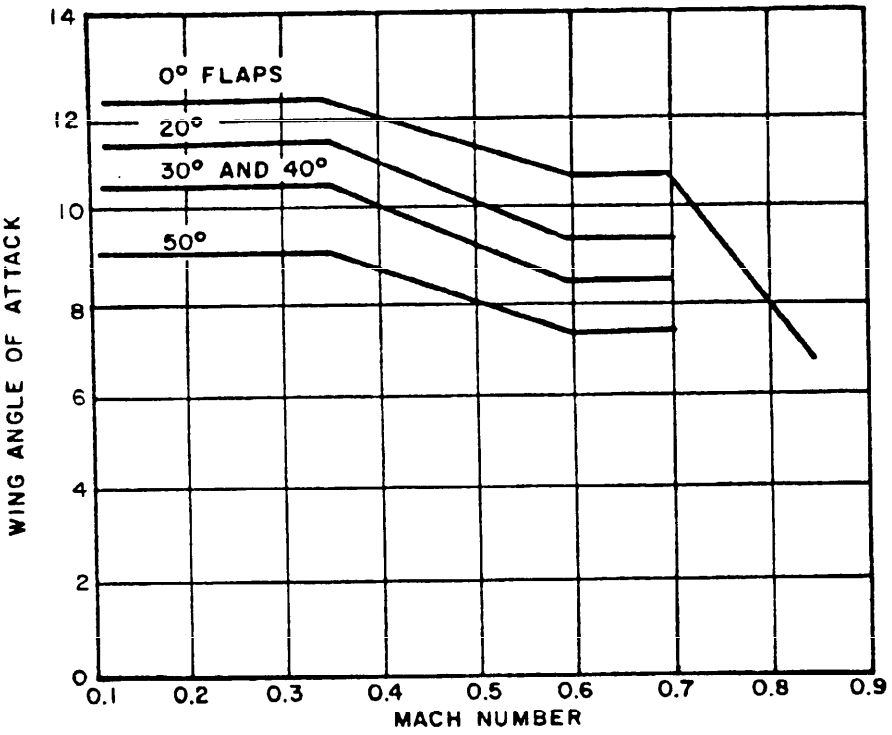
c. The 30° and 40° flap curve shown on figure 2 consists of a straight line from 10.6° α wing to mach 0.35; a straigh line connecting 10.6 α and 0.35 mach to 8.4° and 0.6 mach; a straight line from 8.4° α to 0.7 mach

d. The 50° flap curve shown on figure 2 consists of a straight line from 9.1° α wing to mach 0.35; a straight line connecting 9.1° α and 0.35 mach to 7.3° α and 0.6 mach; a straight line from 7.3° α wing to 0.7 mach.

3.12.2 Angle of attack transmitters. The angle of attack transmitters shall provide angle-of-attack information derived from three-wire synchro output signals from a transmitter conforming to MIL-T-25627. This transmitter is of the low-null electrical-zero type and has a scale factor of 2.7° of transmitter synchro rotation for each degree of sensor vane rotation. The actual rotation of the vane and related wing α versus local α are shown on figure 4.

3.12.3 Airbrake position transmitter. The airbrake position transmitter shall be functionally suitable for the accuracy, reliability, and environmental requirements of this specification. The transmitter shall be of minimum weight and shall have a high reliability rating. The inboard and outboard airbrake systems shall have individual transmitters.

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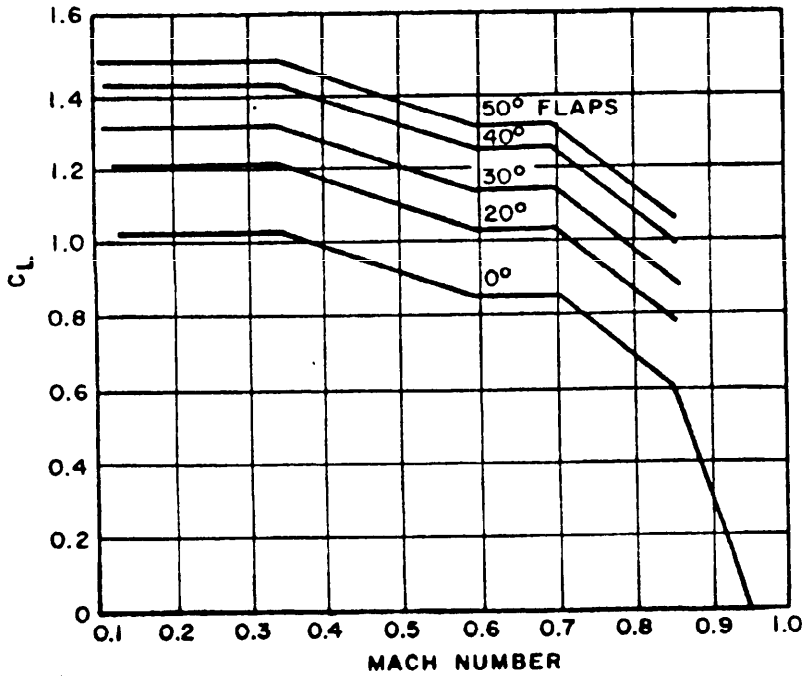


FIGURE 2. Stick Shaker Activation Curve

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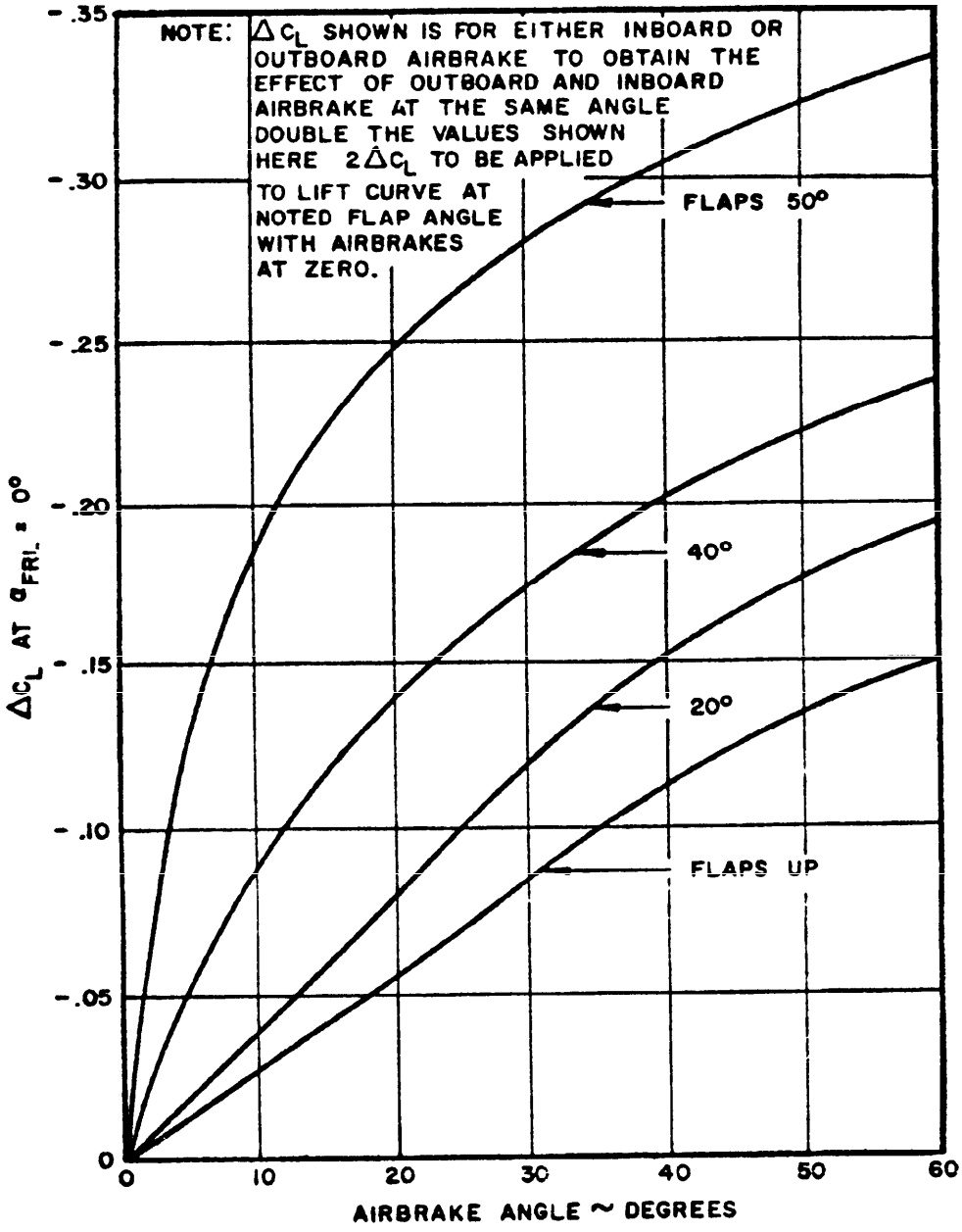


FIGURE 3. Effect of Airbrakes

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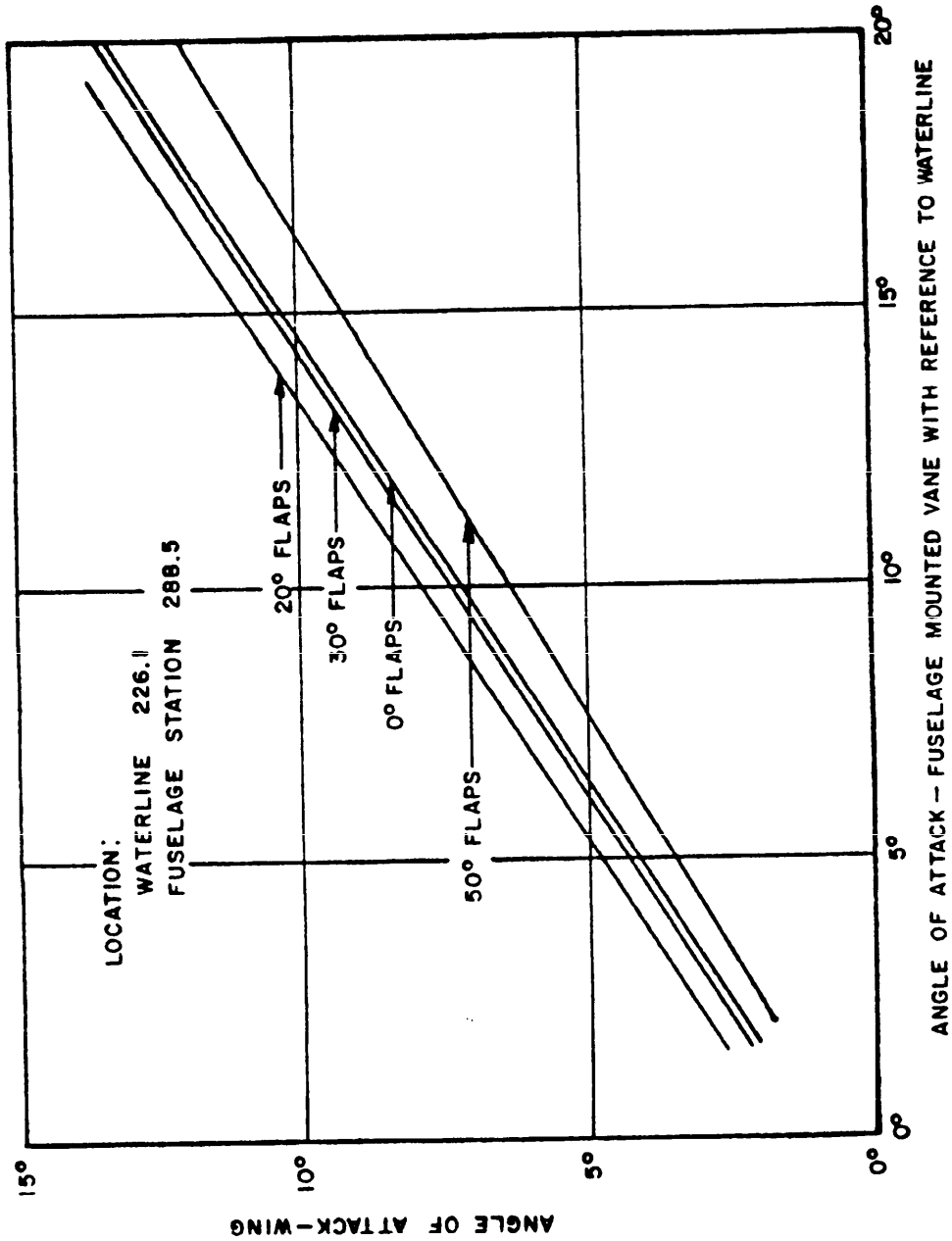


FIGURE 4. α Vane Versus α Wing

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3.12.4 Mach transmitter. The mach transmitter shall be functionally suitable for the accuracy, reliability, and environmental requirements of this specification. The transmitter shall be of minimum weight and shall have a high reliability rating.

3.12.5 Flap position transmitter. The flap position transmitter shall conform to MIL-T-5583.

3.12.6 Stall margin indicator. The stall margin indicator display shall be as shown on figure 5. The indicator shall be integrally lighted in accordance with the color and lighting requirements of MIL-L-25467. The design of the pointer meter movement shall be such that when not energized, the pointer will remain within the stall region of the indicator display. The meter movements shall be electrically isolated from the remainder of the indicator electrical circuitry in such a manner as to preclude interaction problems. Unless otherwise specified herein, under all environmental test conditions and when connected in parallel with a like meter movement, or when connected in parallel with a 1,000-ohm load to a signal source that has an impedance between 350 and 1,000 ohms, the resistance across the meter movement shall be 1,000 ohms ±3 percent at 25°C. The pointer mechanism shall be so damped that there will be not more than 1.5 percent overshoot. The response time of the pointer shall be a maximum of 1/3 second. Response time is the time required for the pointer to reach 90 percent of its final indication upon application of the current. The pointer shall be considered as having come to rest when it has reached within ±1 percent of the scale length of the actual resting point. The accuracy and environmental characterisitcs of the indicator shall be such as to be compatible with the test requirements imposed on the computer. The indicator shall be so designed that when it is revolved about a magnetic compass having a magnetic field intensity of 0.17 to 0.19 oersted, the compass will not deflect more than 1°.

3.12.6.1 Case. The case shall conform to MS33638 for the 2-inch size. Case depth "L" shall not exceed 3.5 inches.

3.12.6.2 Pointer. The pointer shall be in accordance with MS33585-3.

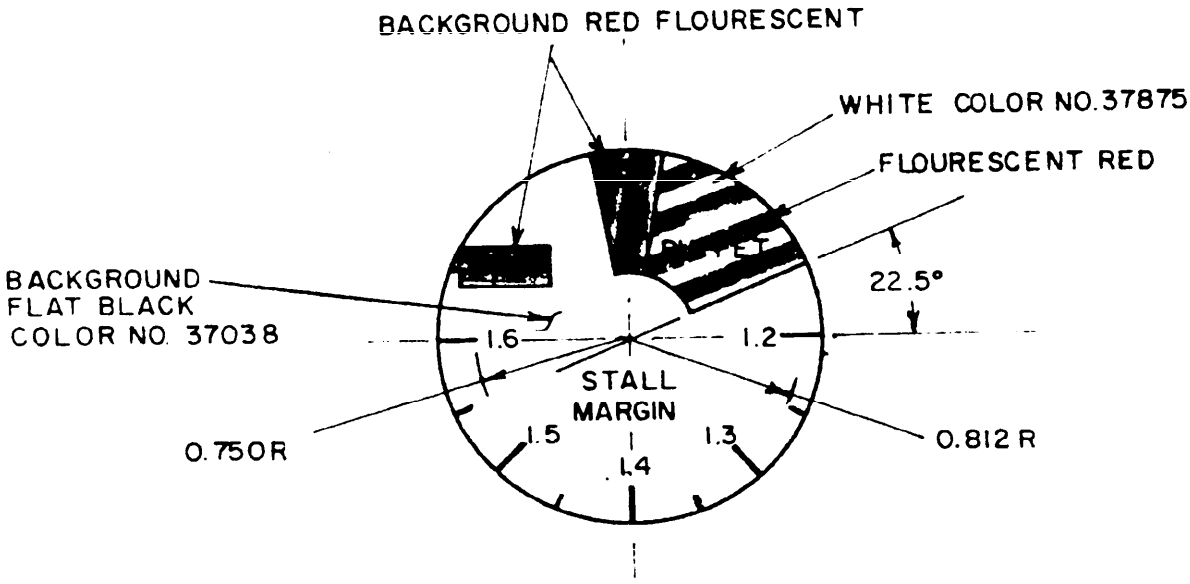
3.12.6.3 Dial markings. Dial markings shall conform to MS33558. The height of the numerals and letters shall be 0.1875 inch except for the word STALL which shall be 0.100 inch.

3.12.6.4 Colors. Unless otherwise specified, colors shall be in accordance with FED-STD-595.

3.12.6.5 Stall margin signals. Stall margin signals shall be in a form compatible with the indicator electrical characteristics and shall be directly proportional to the aircraft's condition relative to stall. The computer shall compare the angle of attack, biased for flaps and mach effect, and airbrakes with the known stall characteristics for the aircraft (see figure 2), and shall provide an output proportional to the margin remaining (V_s) before stall will occur.

3.12.7 Stick shaker. The stick shaker shall meet the environmental, maintainability, and reliability requirements specified herein. When mounted on the aircraft control

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THE WORDS OFF, STALL, AND BUFFET TO BE FLAT BLACK, COLOR NO. 37038. ALL OTHER LETTERING AND NUMBERS TO BE WHITE, COLOR NO. 37875.
RED FLOURESCENT — SWITZER BROS. CO. COLOR NO. 116-1548, OR EQUAL

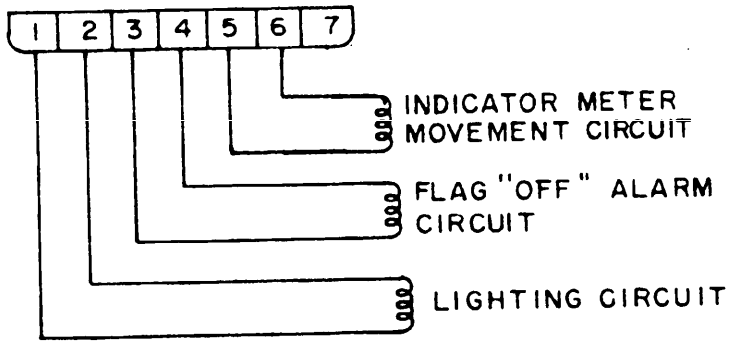


FIGURE 5. Stall Margin Indicator Dial and Wiring Diagram

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column, its output, measured at the wheel, shall be from 12 to 20 cps with an amplitude of from 0.02 to 0.10 inch. In no case shall the induced vibration g-load be in excess of 2g's.

3.12.7.1 Stick shaker activation. When the stall margin reaches the figure 2 specified condition, the computer shall activate the switch that supplies power to the stick shaker motor.

3.12.8 Failure monitors. The system shall be provided with failure monitors which shall be capable of detecting loss or degradation of power, failure of air data transmitters, malfunction of the computer or indicator, and any other malfunction that would degrade the accuracy of the system. When out-of-tolerance conditions exist, the OFF flag on the indicator shall appear.

3.13 Power. Power supplied to the computer shall be controlled by the ON-OFF switch on the control panel and shall be 115V ac 400 cycle and 28V dc. Power of 115V 400 cycle ac shall be supplied directly from a circuit breaker. The 26V 400-cycle a-c power for the angle-of-attack transmitter shall be supplied to the computer for monitoring and distribution. All 400-cycle voltages supplied to the computer shall be of the same phase. Power variation shall be as specified in MIL-STD-704. Power consumption, excluding angle-of-attack heater power, shall not exceed 30w for the computer nor 200w for the stick shaker motor.

3.13.1 Power output. The final contacts that supply power to the stick shaker shall be capable of carrying at least the current required to drive one stick shaker motor.

3.14 Tolerances. The computer shall perform the stall-margin and stick shaker activation computations within ±0.10° of angle-of-attack synchro over a temperature range of -31.7° to +48.9°C. Outside this range, the tolerance limits may be extended linearly with temperature to a maximum of ±0.20° of synchro at the specified temperature extremes. Total system accuracy, including computer, sensors, and indicator, shall not exceed ±0.30° of angle of attack at -31.7° to +48.9°C nor ±0.40° angle of attack at specified extreme temperatures.

3.14.1 The stick-shaker signal and stall-margin sensing circuits shall incorporate filtering if necessary to preclude undesirable susceptibility to interference and instability from internal and external signal oscillations.

3.14.1.1 The stick-shaker sensing circuit shall provide an alarm per figure 2 when α is introduced into the computer at the levels specified for mach and flap positions. The delay in the stick-shaker activation output shall not exceed 0.03 second when the conditions on figure 2 and 3 are satisfied.

3.14.1.2 The stall margin sensing circuit shall provide a continuous signal in accordance with the characteristics of figures 2 and 3. The delay in indicator reading corresponding to a change in flaps, airbrakes, mach, or angle of attack shall not exceed 0.03 second.

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3.15 Vibration isolators. The need for mounting equipment on vibration isolators shall be avoided.

3.16 Interconnection cabling. The equipment shall operate satisfactorily using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when necessary to meet interference control requirements and provided the assembly of the cables to its plugs may be easily accomplished. Twisted wires shall be used where necessary for the prevention of inductive pickup.

3.17 Mounting parts. The component parts shall be mounted in such a manner that they can be readily removed and replaced. Access doors or covers with suitable gaskets shall be provided to affect easy removal of internal parts.

3.18 Fail safety. The circuit design shall be such as to minimize the possibility of unsafe conditions resulting from circuit failure.

3.19 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following additional marking shall be included:

Rating (electrical power supply and maximum servo torque, where applicable).

3.19.1 Connectors. All external electrical connectors shall be legibly and permanently marked by identification numbers.

3.20 Weight. The weight of the computer shall be the minimum permitted by the state of the art and shall not exceed 10 pounds.

3.21 Workmanship. Workmanship shall be of the highest grade consistent with aircraft instrument and electronic manufacturing practice. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking, welding and brazing, painting, riveting and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of computers shall be classified as follows:

a. Preproduction tests

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b. Acceptance tests

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the computer shall be tested in the normal operating position.

4.3.3 Supply voltage. Unless otherwise specified, all tests shall be conducted with 115 ±5V a-c single-phase, 400 ±10-cps power or 28 ±2V d-c power.

4.3.4 Connections. Whenever it is specified that the computer shall be properly connected, it shall be understood that the computer connector terminals are connected to a test fixture that provides the required inputs of angle of attack, mach, and flap position.

4.3.5 Burn-in. Burn-in shall be in accordance with MIL-STD-781.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of three computers representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.1).

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each computer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Stick shaker signal trip level sensitivity

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c. Stall margin signal accuracy

d. Dielectric strength.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Sample computers shall be selected at random in accordance with the following schedule and subjected to the tests listed below:

Production Quantity	No. of Sample Computers
1 to 10	1
11 to 100	2
10 to 200	1
Each additional 200	1 for each 200

a. Individual tests

b. Low temperature operation

c. High temperature operation

d. Extreme voltage and frequency variation

e. Vibration failure

f. Power consumption.

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 computers shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Radio noise interference

c. High temperature exposure

d. Temperature-altitude

e. Thermal shock

f. Sand and dust

g. Rain

h. Fungus

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i. Humidity

j. Salt fog

k. Shock.

4.5.2.2.1 At the option of the procuring activity and with the exception of sampling plan A tests, tests may be divided into 3 groups, as follows, to be conducted on 3 sets of computers if a need exists to expedite testing. Each computer shall meet the requirements of all the individual tests after completing the group of specific tests to which submitted. At the option of the procuring activity, group II testing may be conducted on empty cases which have all external parts installed or attached as on a completed computer.

GROUP I	GROUP II	GROUP III
Radio noise interference	Sand and dust	Shock
High temperature exposures	Rain	Reliability
Temperature-altitude	Fungus	
Thermal shock	Humidity	
	Salt fog	

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.2.4 Sampling plan C. Unless otherwise specified, three computers shall be selected at random from the first 20 computers of each production quantity of 100 and subjected to the test specified in 4.6.13. Rejection shall be as specified therein.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The computer shall be inspected to determine compliance with the requirements herein with respect to materials, workmanship, marking, and design.

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4.6.2 Stick shaker signal trip level sensitivity. Various combinations of angle-of-attack, airbrakes, flap-position, and mach signals shall be introduced into the computer. In all cases, the stick-shaker circuit shall trigger when the conditions specified in 3.14 are satisfied.

4.6.3 Stall margin signal accuracy. With the stall margin indicator connected to the computer, various combinations of angle-of-attack, airbrakes, flap-position, and mach signals shall be introduced into the computer. In all cases, the stall margin indicator shall display the correct values of stall margin within the tolerances specified in 3.14.

4.6.4 Dielectric strength. A potential of 500V ac shall be applied between isolated pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.5 Low temperature operation. The computer shall be properly connected with no power applied and subjected to a temperature of -54° ±2°C for a period of 4 hours. At the end of this period and while still at the low temperature, power shall be applied. The computer shall then meet the individual tests.

4.6.6 High temperature operation. The computer shall be properly connected, with no power applied, placed in a chamber, and maintained at a temperature of 71° ±2°C for a period of 4 hours. While still at the high temperature, power shall be applied and the computer shall meet all individual tests. The temperature shall be increased to 98°C for a period of 10 minutes with the computer operating. The computer shall again meet the individual tests.

4.6.7 Extreme voltage and frequency variation. The computer shall be properly connected and, with input power applied, shall meet the individual tests at all combinations of 105V 380 cps and 125V 420 cps. With input power applied, the computer shall then meet the individual tests at all combinations of 105V 420 cps and 125V 380 cps, except that the total system tolerances shall be within ±0.40° angle of attack for the extreme voltage and frequency variations. The tolerance allowed by MIL-STD-704 shall be applicable for 28V d-c power.

4.6.8 Vibration failure. The computer shall be properly connected with power applied and subjected to a vibration test in accordance with MIL-STD-810, method 514, IBIC. At the end of this test, the computer shall meet the individual tests. The computer shall then be subjected to a vibration of 0.4 inch double amplitude or 1.0g (whichever is less) within the frequency range of 1 to 10 cps for a period of 15 minutes. There shall be no transients or discontinuity in outputs for any mode of operation. The computer shall then meet the individual tests. No damage to the computer shall result from this test.

4.6.9 Power consumption. The computer shall be properly connected with power applied. The a-c and d-c power consumption for each mode shall not exceed the values specified in 3.13. The computer shall meet the individual tests.

4.6.10 Radio noise interference. The radio noise interference test shall be conducted in accordance with MIL-I-6181 over the frequency range of 150 kc to 150 mc.

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4.6.11 Environmental tests. The computer shall be subjected to the following tests, conducted in accordance with the specified methods of MIL-STD-810:

4.6.11.1 High temperature exposure. This test shall be conducted in accordance with method 501. The period of exposure shall be 24 hours. The computer shall be properly connected, with no power applied during exposure. At the end of the 24-hour period and while still at the high temperature, power shall be applied for 1 hour. The computer shall meet the individual tests specified herein. The computer shall operate without sticking or erratic performance during this test.

4.6.11.2 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with method 504 for class 2 equipment, except that the altitude shall be 80,000 feet. The computer shall be connected with no power applied. At the end of the 72-hour period, pressure in the chamber shall be reduced to the equivalent of 80,000 ±500 feet, and power applied. The computer shall operate immediately and its operation shall be observed for 1 hour. There shall be no evidence of erratic performance. All mode relays and rate circuits shall be tested. Pressure in the test chamber shall be increased to atmospheric and, after the computer has returned to room temperature, it shall meet the individual tests.

4.6.11.3 Thermal shock. The computer shall be subjected to a thermal shock test in accordance with method 503. After the computer has returned to room temperature, connectors and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The computer shall then meet the individual tests.

4.6.11.4 Sand and dust. The sand and dust test shall be conducted in accordance with method 510. Upon completion of this test, the computer shall be examined to determine that no sand and dust has entered the computer case. The computer shall then meet the individual tests.

4.6.11.5 Rain. The rain test shall be conducted in accordance with method 506. The computer shall then meet the individual tests.

4.6.11.6 Fungus. The fungus test shall be conducted in accordance with method 508, procedure I. Upon completion of this test, the computer shall meet the individual tests. There shall be no deterioration nor shall any part of the computer support fungus growth.

4.6.11.7 Humidity. The computer shall be subjected to a humidity test in accordance with method 507 after which it shall meet the individual tests. There shall be no evidence of corrosion or rust which will affect subsequent operation.

4.6.11.8 Salt fog. The computer shall be subjected to a salt fog test in accordance with method 509. At the end of the 48-hour period, the computer shall meet the individual tests.

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4.6.11.9 Shock. The computer shall be subjected to a shock test in accordance with method 516, procedure I. The equipment shall not be damaged nor subsequently fail as a result of this test. The computer shall then meet the individual tests.

4.6.12 Reliability

4.6.12.1 Preproduction. The computers shall be subjected to a reliability test in accordance with MIL-STD-781 which shall demonstrate a minimum MTBF of 1,000 hours. Each of the three computers shall be operated for a minimum of 3,100 hours for a total test time of 9,300 hours for an accept decision. Accept-reject criteria shall be in accordance with test plan I and the test environment shall be in accordance with test level E. Reporting shall also be in accordance with MIL-STD-781. At 100-hour intervals, individual tests shall be run. Failure to meet the individual tests shall be counted as a failure. All inputs shall be applied to the computer at their maximum value during power-on time. The output loads shall be equivalent to the indicator and switching loads. If a reject decision is reached due to 17 failures during the 9,300 hours of testing, the contractor shall modify the equipment to eliminate the cause of failure and continue the test until an accept decision is reached. If a continue-test decision is reached at the end of the test period, the contractor shall submit all test data to the procuring activity for analysis and shall continue the test until an accept or reject decision is reached. The contractor may incorporate corrective action during any portion of the test provided such action is first approved by the procuring activity. Subsequent to this corrective action, all equipment produced on the contract shall be modified by the contractor to the approved configuration.

4.6.12.2 Acceptance. An acceptance reliability test shall be conducted to demonstrate a minimum MTBF of 1,000 hours. Acceptance shall be in accordance with test plan I and the test environment shall be in accordance with test level E. During this test, all inputs shall be applied to the computer at their maximum value. The outputs shall be equivalent to the indicator and switching loads. If a reject decision is reached, the contractor shall modify the equipment to eliminate the cause of failure and continue the test until an accept decision is reached. The contractor may incorporate corrective action during any portion of the test provided such action is first submitted to the procuring activity for analysis and approval. Subsequent to this corrective action, the contractor shall modify all computers to the approved configuration.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The CPU-81/A computer covered by this specification is intended for use in the C/KC-135 aircraft for integrating angle-of-attack, airbrakes, flap-position, and mach data and to provide output signals proportional to the aircraft's condition relative to stall.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B and C tests will not be conducted

c. Selection of applicable levels of packaging and packing and reinspection date markings.

6.2.1 Test report. The contractor will submit three complete copies of the test report (see 4.4.2) to the procuring activity.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 11, 71, 84

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0270

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